Matrix inequalities and norms

The maximum permanent on a positive-semidefinite unitary orbit

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Provenance: explicit exact extremal problem; no conjectured optimizer is supplied

Rating rationale: An exact formula for arbitrary spectra would resolve a longstanding permanent optimization barrier and inform prescribed-spectrum matrix optimization.

Exact formula for one-exceptional-eigenvalue spectra — 12 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. Submission and verified affiliation.

The Theorem and its proof determine the maximum for every spectrum $(\alpha, \beta, \dots, \beta)$ with $\alpha, \beta \geq 0$ in every order, including $\alpha < \beta$. It compares finitely many explicit support-size polynomials and supplies maximizers. The formula for arbitrary spectra remains open.

A separate independent Codex AI-agent audit passed this limited scope. This is informal automated review; no complete resolution, historical novelty, external human peer review or formal verification is claimed. No Lean verification was performed.

Problem statement

For every integer $n \geq 1$ and every real list $\lambda = (\lambda_1, \dots, \lambda_n)$ with $\lambda_i \geq 0$, determine the exact value of

$$M_n(\lambda) = \max_{U \in \mathcal{U}_n} \operatorname{per}(U^* \operatorname{diag}(\lambda_1, \dots, \lambda_n) U), \qquad \mathcal{U}_n = \{U \in \mathbb{C}^{n \times n} : U^* U = I_n\},$$

where $\operatorname{per}(C) = \sum_{\sigma \in S_n} \prod_{i=1}^n c_{i,\sigma(i)}$ and S_n is the permutation group. The permanent of these Hermitian positive-semidefinite matrices is real. Compactness ensures the maximum exists. A resolution should express its value from the prescribed eigenvalues for arbitrary order, rather than only bound the objective or restate its optimization definition.

Relevance

The eigenvalues fix an entire unitary similarity class while the permanent varies within it. Determining its extreme value is a concrete optimization problem over matrices with a prescribed spectrum.

References

1. F. Zhang, *An update on a few permanent conjectures*, Special Matrices 4 (2016), 305–316, passage headed "Marcus–Minc max-per-U problem 1965." Primary text; journal.
2. J. H. Drew and C. R. Johnson, *Counterexample to a conjecture of Mehta regarding permanental maximization*, Linear and Multilinear Algebra 25(3) (1989), 253–254, the counterexample. DOI.

Status check — 2026-09-10

Zhang's latest arXiv record is v1 and explicitly lists the general maximization problem. Its statement and discussion of the equal-diagonal counterexample were inspected. The original 1989 full text was not retrieved; that counterexample's role was checked through Zhang's explicit account and bibliography. Searches used *maximum permanent unitary 2026*, *maximum permanent eigenvalues 2025 2026*, and *Marcus-Minc max-per-U problem*. No general spectral formula was located. The claim that an equal-diagonal representative always maximizes the permanent is false and is not imposed. The other catalog permanent entries concern inequalities between distinct matrix functions or products, not this exact orbit maximum.

Audit update (2026-09-10): Rechecked Zhang's Marcus–Minc max-per-U statement and searched for later exact spectral formulas. Equal-diagonal optimality remains an invalid shortcut; no all-spectrum determination was located. This is a bounded literature check, not a proof that no solution exists.